SS3 to opal: IO yellowfin with age-specific K

Stock Synthesis GrowthModel 3 growth in opal, with SS3 and opal-estimated selectivity

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Table of contents

1 Overview 1
2 SS3 growth in opal 2
3 Configurations 3
3.1 Selectivity in fit C 4
4 Conversion checks 4
5 Optimisation 5
6 Index fits 6
7 Composition fits 7
8 Selectivity 8
8.1 Fit C2: the prior-dominated parameters fixed 10
9 Biomass and depletion 11
9.1 Against SS3 and against the truth 12
10 Remaining differences from SS3 12
11 Credit 13
12 Session 13

1 Overview

This vignette follows *SS3 to opal: IO yellowfin*, which converted the Indian Ocean yellowfin SS3 model into opal and compared fits. The SS3 model, the baseline here, was built by the IATTC analyst team (Mark N. Maunder, Haikun Xu and Carolina Minte-Vera) for the Spatial Stock Assessment Simulation Experiment; see **Credit**. That comparison found that fitted opal models diverged from SS3 because of growth. SS3 uses `GrowthModel 3`, with age-specific K, and opal's single-k Schnute curve could not represent it. It was 5 to 16 cm too long at ages 4–8, which shifted every predicted length composition. A local patch supplying SS3's length-at-age confirmed the diagnosis.

opal 0.1.1 (`dev` branch, commit `c493461`) implements age-specific K as an option to `get_growth()`. **The method is Stock Synthesis'**: cumulative K multipliers, Linf derived from the oldest-age K, and SS3's plus-group length approximation (Methot and Wetzel 2013; SS3 source, `SS_Label_Info_16.2.2–16.2.4.1.4`). This vignette uses it with the growth parameters read from the SS3 files, so no patch is needed. It then goes one step further than the first vignette, and lets opal estimate its own selectivity.

Key results.

- opal's length-at-age now equals SS3's to Report.sso print precision: max relative error 4.6e-06 over ages 1–28.
- **A1**, with SS3's selectivity: B0 is 1.03 × SS3's and terminal depletion is 0.404, against SS3's 0.399. With Schnute growth the same configuration gave 0.77 × and 0.267.

- **A2**, with control-file biology through opal's PLA, now sits close to A1: B0 1.03 × SS3's (with Schnute growth, 0.43 ×).
- **C** estimates the opal equivalents of the 58 selectivity parameters SS3 estimated. B0 is 0.88 × SS3's and terminal depletion 0.307. Opal check: passed. Nine of those parameters are set by opal's selectivity prior rather than the data.
- **C2** fixes those nine at their seeded values and estimates the other 49: B0 0.88 × SS3's, terminal depletion 0.307. Opal check: passed.
- Against the simulated truth (terminal depletion 0.468), see section 8.

Step	File	Does
1	`code/rtmb/scripts/01-fit-io-yft-ss3-opal.r`	Earlier fits; supplies the SS3 reference quantities and the Schnute-growth reference lines
2	`code/rtmb/scripts/02-extract-io-yft-sim-truth.r`	Simulated truth cache
3	`code/rtmb/scripts/03-fit-io-yft-age-k.r`	The fits in this vignette; writes `vignettes/data/io-yft-age-k-cache.rds`
4	`code/rtmb/vignettes/ss3-to-opal-io-yft-age-k.qmd`	This document; reads only the caches

The cache was written 2026-10-13 22:21:41 UTC with R 4.5.3, opal 0.1.1 and r4ss 1.56.0.

2 SS3 growth in opal

With `growth.model: ss3_age_k` in the converter config, every growth input is read from the SS3 files. Nothing is fitted.

input	value
L1 (L_at_Amin, age 1)	22.0
L2 (L_at_Amax, age 28)	145.0
k per quarter (VonBert_K / 4)	0.11375
K multiplier ages	2, 3, 4, 5, 6, 7, 8, 9, 10
K multipliers	0.5, 0.75, 1, 1, 1, 1.8, 1.8, 1.2, 1.2
CV1, CV2 (CV_young, CV_old)	0.10, 0.10
plus-group adjustment	yes

SS3 reads K as an annual rate and applies it over the 3-month season, so opal's per-step k is K / 4. M is scaled the same way. The multipliers are cumulative: K at an age is the previous age's K times that age's multiplier.

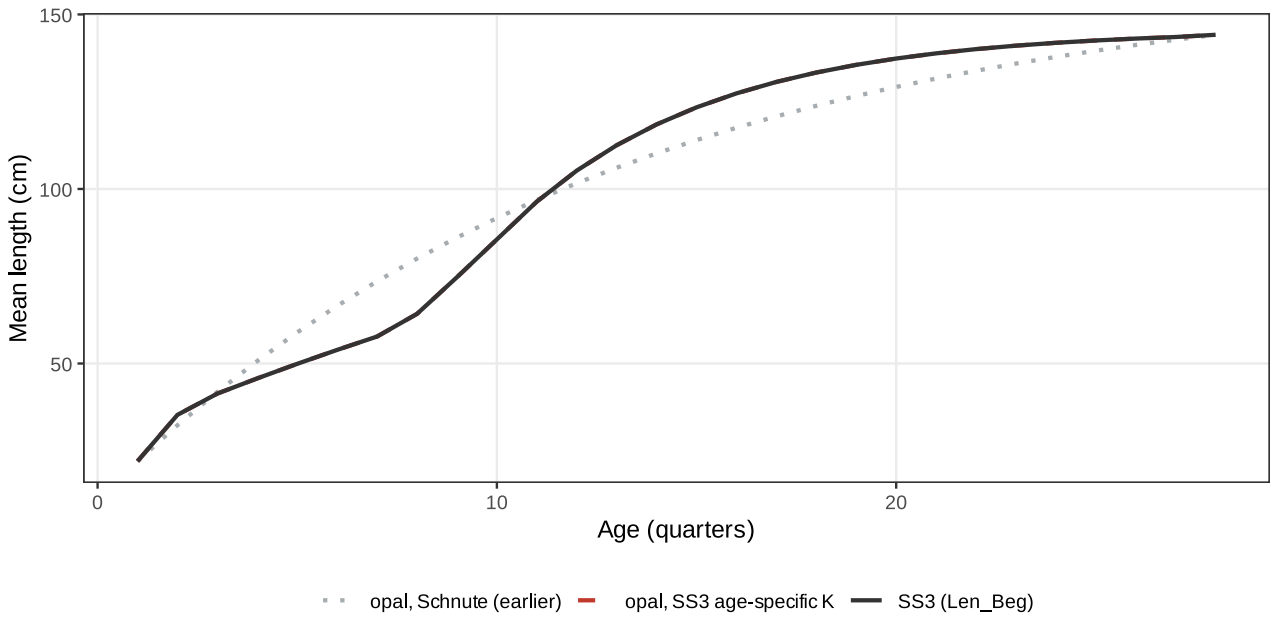


Figure 1: Mean length-at-age: SS3, opal with SS3 age-specific K, and the single-k Schnute curve used previously.

Age	SS3 Len_Beg (cm)	opal (cm)	opal − SS3 (cm)
1	22.0000	22.0000	0.00000
2	35.2865	35.2865	−0.00005
3	41.3840	41.3840	−0.00003
4	45.7348	45.7348	0.00003
5	49.9040	49.9040	0.00001
6	53.8991	53.8991	−0.00002
7	57.7273	57.7273	0.00002
8	64.2198	64.2198	−0.00001
9	74.7210	74.7210	−0.00004
10	85.5489	85.5489	0.00002
15	123.3830	123.3827	−0.00034
20	137.3690	137.3695	0.00050
25	142.5400	142.5403	0.00032
28	144.1680	144.1678	−0.00019

The SD of length-at-age is CV × length with CV1 = CV2 = 0.1, the same as SS3's `SD_Beg`. With growth exact, opal's probability-of-length-at-age matrix (PLA) is SS3's age-length key at the start of the timestep.

3 Configurations

Fit	Biology	Selectivity	Estimated
A1	endgrowth (SS3 age vectors)	SS3 selectivity-at-age, fixed	log B0, 256 rec devs, 4 q

Fit	Biology	Selectivity	Estimated
A2	control file, through the PLA	SS3 selectivity-at-age, fixed	as A1
C	endgrowth	opal length-based curves	as A1, plus 58 selectivity parameters
C2	endgrowth	as C	as A1, plus 49 selectivity parameters (nine of C's held at seeded values)

All four fix growth at SS3's values, and steepness and σ_R at SS3's. They start from the converted SS3 optimum. The length compositions enter as multinomials with SS3's `mult_by_lencomp_N` variance adjustments.

3.1 Selectivity in fit C

Fit C estimates the opal equivalents of exactly the selectivity parameters SS3 estimated (positive phase in `control.ss_new`):

SS3 fleet	SS3 pattern	SS3 estimated	SS3 fixed	opal equivalent
1–14	20 (double-normal)	P1, P2, P3, P4	P5 = −6; P6 = −999	double-normal; columns 1–4 estimated, 5–6 at seeded values
15, 16	15 (mirror fleet 14)	—	—	share fishery 14's parameters
17 (survey)	12 (logistic)	P1, P2		logistic; both estimated, shared by opal index fisheries 17–20

SS3's double normal (pattern 20) and opal's take their six parameters in the same order, so SS3's P1–P4 map to opal columns 1–4. The initial and final values (columns 5–6) stay at their seeded values. opal has no equivalent of SS3's `-999` final-value option, so the seeded value stands in for it. Starting values come from fitting each opal curve to SS3's selectivity projected onto length (seeding RMSE ≤ 0.050), within opal's default bounds of [−7, 7]. opal's default N(0, 2) prior on `par_sel` applies. The survey curve has less information in opal than in SS3, because SS3 also fitted the survey's generalised size-composition data, which the converter omits.

4 Conversion checks

quantity	rel_diff	tol	pass
CPUE observation count	0.00e+00	1e-06	TRUE
LF total input sample size	0.00e+00	1e-06	TRUE
LF observation rows	0.00e+00	1e-06	TRUE
unfished spawning potential per recruit	4.69e-05	1e-04	TRUE
total weight summed over ages	0.00e+00	1e-06	TRUE
mean length-at-age, SS3 age-specific K (max relative error)	4.60e-06	5e-05	TRUE

The 16 catch totals reconcile exactly too (not shown). The growth row is new: with SS3's own growth form, length-at-age is now an exact check rather than an approximation. The SPR0 tolerance allows for SS3's truncated plus-group equilibrium, as explained in the first vignette.

5 Optimisation

As before, **every fit starts from the converted SS3 optimum, so these times are for refining a solution that is already close**. Fit C also starts from seeded selectivity curves.

Stage	A1 (s)	A2 (s)	C (s)	C2 (s)
convert SS3 files	10.1	5.7	—	—
build opal objects	35.0	33.4	30.6	31.2
opal_fit: tape + nlminb (2 passes)	39.7	65.9	47.8	52.6
rebuild tape	15.2	17.1	16.7	16.6
Newton polish	17.5	17.6	25.2	24.7
attach fit + opal_check	41.5	43.1	47.0	47.2
seed selectivity	—	—	4.2	4.2
total	159.0	182.8	171.5	176.5

Fit	Pass	Objective	Convergence	Iterations	Function evals	Gradient evals
A1	1	14214.61	0	265	288	266
A1	2	14214.61	0	19	24	20
A2	1	14214.61	0	1073	1115	1074
A2	2	14214.61	0	64	74	65
C	1	14393.33	0	366	389	366
C	2	14393.33	0	38	44	39
C2	1	14406.58	0	528	558	528
C2	2	14406.58	0	49	56	50

Fit	Parameters	Selectivity parameters	Objective, start	Objective, fit
A1	261	0	14242.1	14214.6
A2	261	0	14242.0	14214.6
C	319	58	14920.6	14393.3
C2	310	49	14920.6	14406.6

Fit	Max |grad|, nlminb	Max |grad|, final	Newton steps	opal check	Estimable
A1	3.6e-03	7.6e-12	2	passed	all
A2	9.1e-04	8.8e-12	2	passed	all
C	3.4e-03	1.1e-11	2	passed	all
C2	5.3e-03	8.0e-12	2	passed	all

nlminb runs two passes. A few Newton steps then take the maximum gradient below 1e-8, and `opal_check()` runs at a gradient tolerance of 1e-4, checking the Hessian, bounds, population and estimability. Timings are from a Intel(R) Xeon(R) Processor @ 2.80GHz machine with 4 cores. The first vignette found wall-clock times on this cloud container varying by up to 2× between sessions with identical iteration counts, so treat the times as indicative.

6 Index fits

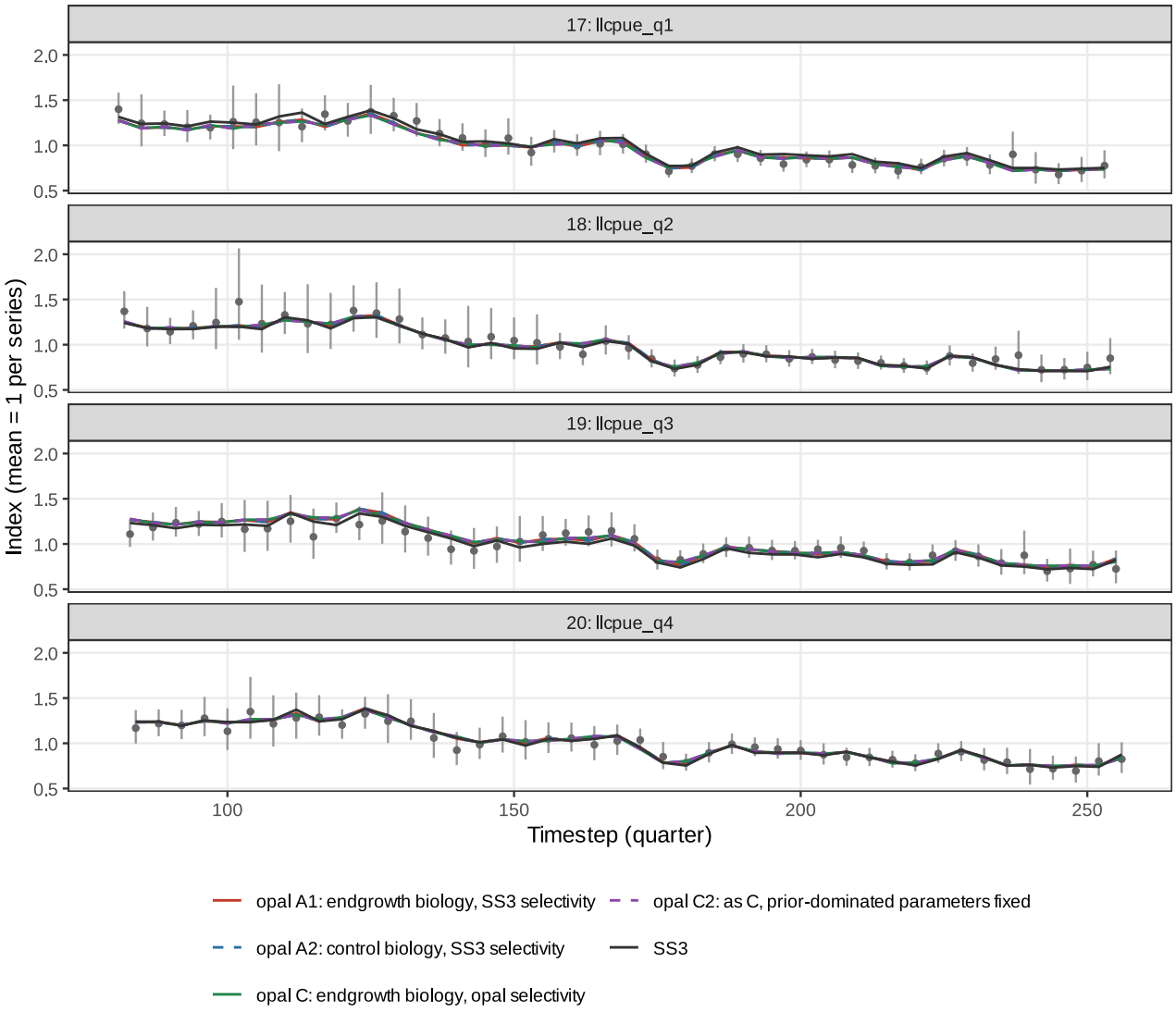


Figure 2: Observed index (points, ±1.96 SE) with the fitted opal predictions and SS3's expected values, each quarterly series on a mean-one scale.

Model	RMSE of log residuals
opal A1: endgrowth biology, SS3 selectivity	0.0616
opal A2: control biology, SS3 selectivity	0.0616
opal C: endgrowth biology, opal selectivity	0.0621
opal C2: as C, prior-dominated parameters fixed	0.0621
SS3	0.0646

7 Composition fits

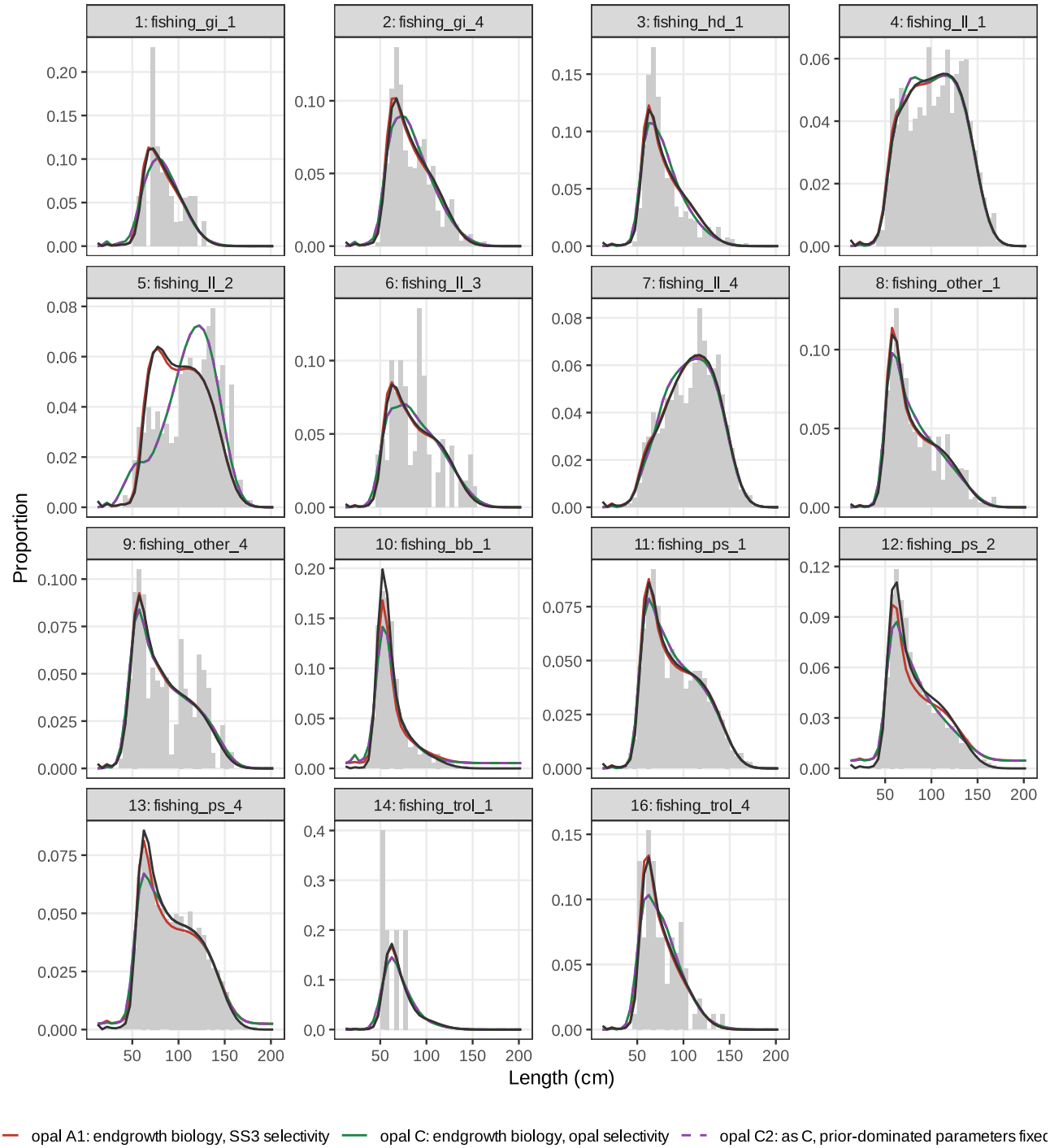


Figure 3: Length compositions aggregated over all observations per fishery, weighted by effective sample size.

With SS3's growth, A1's predicted compositions sit on SS3's expectation; in the first vignette, the Schnute curve pushed them to the right. Fit C's predictions differ from SS3's only through selectivity.

Model	NLL	length comps	NLL index	Recruitment penalty	Priors
opal A1: endgrowth biology, SS3 selectivity		14253.9	−240.2	135.1	239.7
opal A2: control biology, SS3 selectivity		14253.9	−240.2	135.1	239.7
opal C: endgrowth biology, opal selectivity		14236.2	−240.3	136.3	431.1

Model	NLL	length comps	NLL index	Recruitment penalty	Priors
opal C2: as C, prior-dominated parameters fixed	14236.2	−240.3	136.3	444.4	

The composition NLL is not comparable with SS3′s, because opal's multinomial keeps its normalising constant.

8 Selectivity

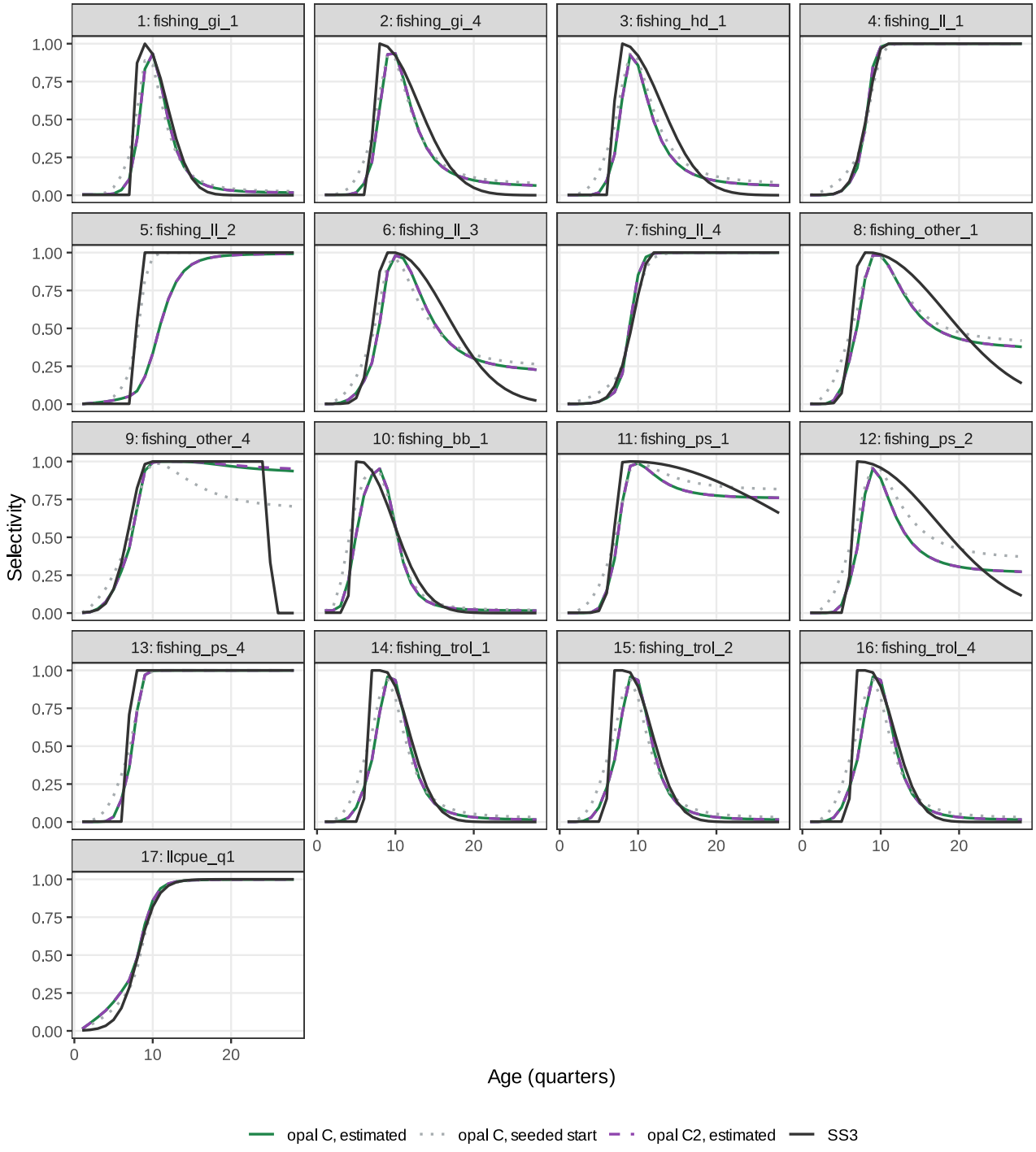


Figure 4: Selectivity-at-age: SS3 (A1 and A2 use it unchanged), the seeded starting curves shared by C and C2, and the C and C2 estimates.

Panels 15 and 16 mirror 14, and 17 stands for all four quarterly index series (18–20 are identical). The seeded curves are SS3′s selectivity projected onto length and fitted with opal's length-based families. The estimated curves show where the length compositions, read through opal's dynamics, pull them.

Fishery	Peak	Plateau	Asc. width	Desc. width	a50	Width
1	−0.60	−2.39	−3.99	−2.23	—	—
2	−0.70	−2.50	−4.62	−1.25	—	—
3	−0.75	−3.61	−5.54	−0.88	—	—
4	−0.56	0.01	−3.15	0.00	—	—
5	0.38	0.00	−0.68	0.00	—	—
6	−0.55	−1.84	−2.69	−1.34	—	—
7	−0.32	0.00	−2.29	0.00	—	—
8	−0.77	−2.00	−4.24	−0.58	—	—
9	−0.61	−0.13	−2.59	−0.03	—	—
10	−0.96	−3.51	−5.76	−1.20	—	—
11	−0.77	−2.08	−5.25	−0.89	—	—
12	−0.79	−3.90	−5.20	−0.52	—	—
13	−0.76	0.01	−5.19	0.00	—	—
14	−0.68	−2.20	−3.42	−1.99	—	—
17	—	—	—	—	−0.74	−0.67

The table gives fitted values on opal's parameter scale (positions in units of sd(len_mid) about mean(len_mid), widths on the log scale). None sits at opal's default bound of ±7.

Several estimates are set by the prior, not the data. opal places a N(0, 2) prior on every `par_sel` value. A parameter that the length compositions cannot inform settles at the prior mean of 0.

Fishery	Parameter	Seeded start	Estimate
4: fishing_ll_1	plateau	3.462	0.007
5: fishing_ll_2	plateau	3.551	0.000
7: fishing_ll_4	plateau	3.217	0.005
13: fishing_ps_4	plateau	3.642	0.014
4: fishing_ll_1	descending width	−3.787	0.001
5: fishing_ll_2	descending width	−3.804	0.000
7: fishing_ll_4	descending width	−3.730	0.001
9: fishing_other_4	descending width	0.615	−0.034
13: fishing_ps_4	descending width	−3.841	0.002

These 9 parameters moved from their seeded values to within 0.05 of zero. For the asymptotic fisheries (4, 5, 7, 13) they are the plateau and descending width, the whole right-hand side of the curve. Selectivity there has reached 1 before the largest lengths observed, so the data say nothing about where a descending limb would start. Fishery 9′s descending width goes the same way, and its plateau moves from −2.55 to −0.13, which is why its curve stays flat and then drops at the largest ages. SS3 estimated the corresponding parameters too,

without a prior. In opal they are estimable (the Hessian is positive definite and `opal_check()` passes) only because of the prior.

The priors also explain why fit C has a higher total objective than A1 despite fitting the data better. They add 191 units relative to A1, more than the 179-unit difference in total. Without them, C's composition NLL is 17.7 lower than A1′s, with the index and recruitment terms almost unchanged.

The survey logistic (fishery 17) also barely moved from its seeded start, since opal's only information on it is the index itself.

8.1 Fit C2: the prior-dominated parameters fixed

Fit C2 repeats C with those nine parameters held at their seeded values, the opal curves fitted to SS3′s selectivity, and estimates the other 49.

Quantity	C	C2
Objective	14,393.325	14,406.585
NLL length comps	14,236.177	14,236.176
NLL index	−240.255	−240.255
Recruitment penalty	136.283	136.283
Priors	431.110	444.369
B0 (t)	4,095,727.343	4,095,646.518
Terminal SSB/B0	0.307	0.307

C2 is the same fit as C. The likelihood components agree to the third decimal, B0 differs by 0.002%, and the other estimated selectivity parameters move by at most 0.024. The selectivity curves differ by at most 0.015 (fishery 9 at age 28, the plus group) and by under 1e-4 everywhere else. The only difference in the objective is the prior: opal's N(0, 2) prior still applies to the fixed values, which adds a constant 13.26 units, exactly the 13.26 difference in the prior term.

This confirms that those nine parameters are unidentified by the data. The likelihood is flat in them, so it makes no difference whether the prior or a fixed value sets them. It also shows that C's lower stock size (B0 0.88 × SS3′s, against A1′s 1.03 ×) comes from the selectivity parameters the length compositions *do* inform, not from the prior. Fixing the unidentified parameters is the cleaner configuration: the selectivity parameters in C2 are all informed by the data, and the result does not depend on the prior.

9 Biomass and depletion

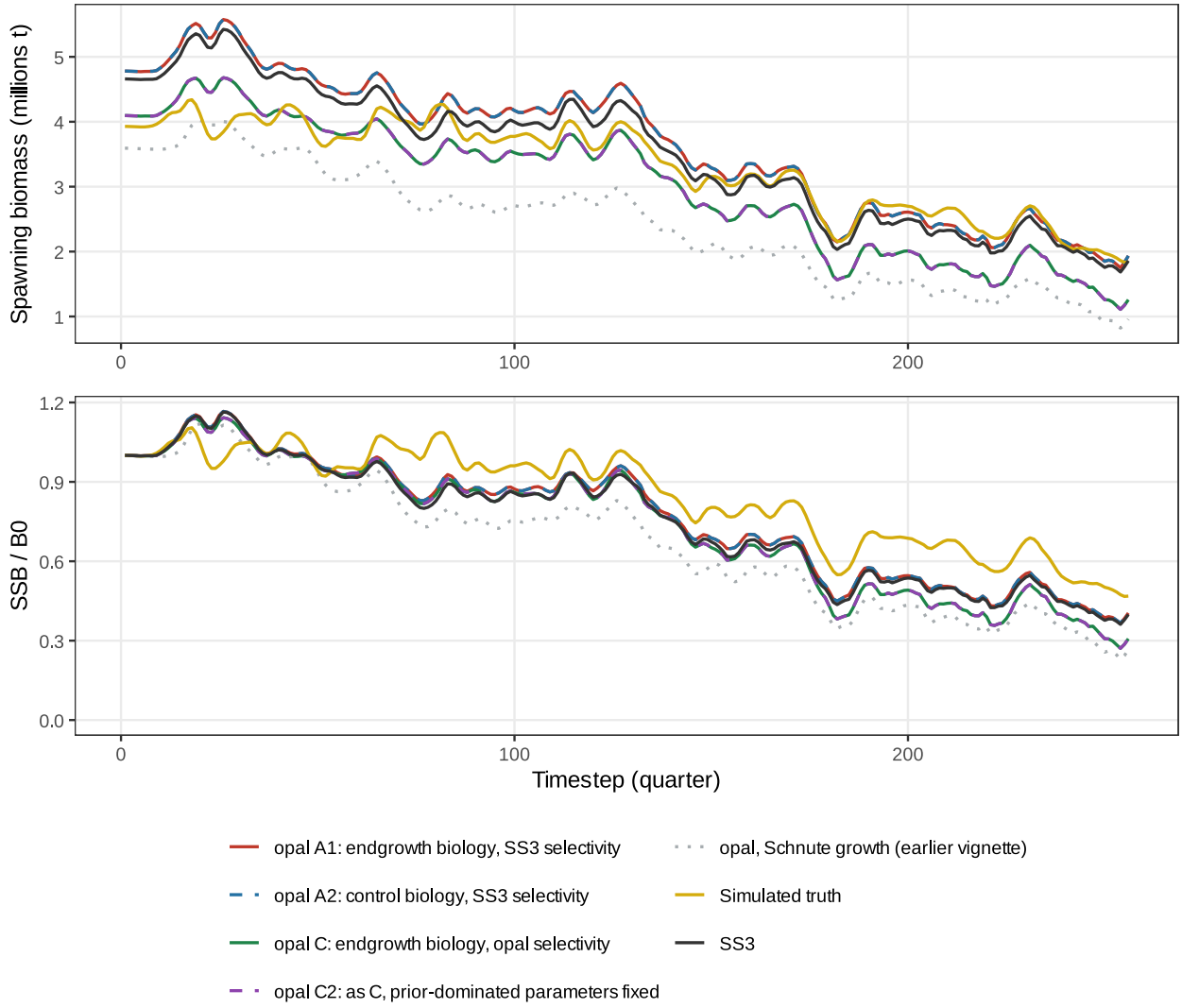


Figure 5: Spawning biomass (top) and static depletion SSB/B0 (bottom): simulated truth, SS3, the four opal fits, and the earlier opal fit with Schnute growth.

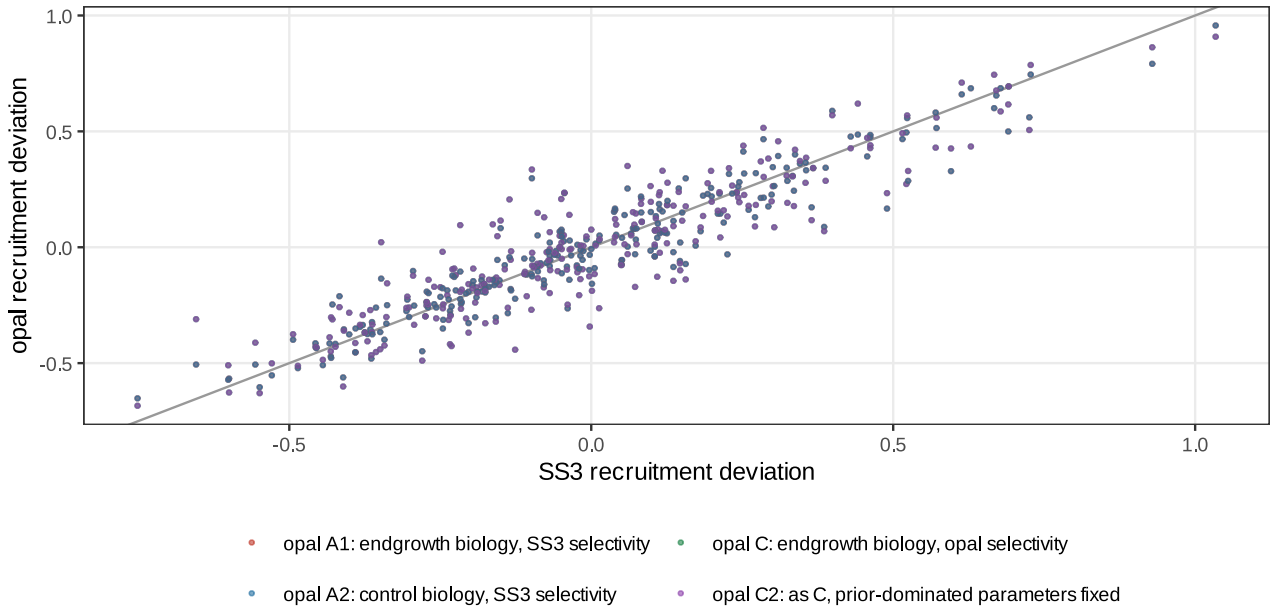


Figure 6: Recruitment deviations: opal fits against SS3's. opal's deviation for timestep t sets the age-1 recruits entering at t + 1.

9.1 Against SS3 and against the truth

Model	B0 / SS3 B0	B0 / true B0	SSB mean |% diff| from SS3	SSB mean |% diff| from truth	Terminal SSB/B0	Max |depletion error| vs truth
opal A1: endgrowth biology, SS3 selectivity	1.03	1.22	4.6	11.0	0.404	0.214
opal A2: control biology, SS3 selectivity	1.03	1.22	4.7	11.1	0.405	0.214
opal C: endgrowth biology, opal selectivity	0.88	1.04	15.5	14.6	0.307	0.243
opal C2: as C, prior-dominated parameters fixed	0.88	1.04	15.5	14.6	0.307	0.243
opal, Schnute growth (earlier vignette)	0.77	0.91	33.0	29.9	0.267	0.326
SS3	1.00	1.19	—	9.8	0.399	0.252
Simulated truth	0.84	1.00	—	—	0.468	—

10 Remaining differences from SS3

With growth reproduced, the structural differences between these opal fits and SS3 are the ones measured in the first vignette:

- **Catch timing.** opal removes each quarter's catch at the start of the step; SS3 uses Baranov catch through the step (`F_Method = 3`). At SS3's parameter values this alone moves opal's SSB by up to 10%.

- **Spawner-recruit timing.** opal's first age class at t + 1 comes from SSB at t + 1; SS3's age-0 recruits at t come from SSB at t.
- **Index.** opal fits the longline index as four quarterly series with four catchabilities, against abundance at the start of the step. SS3 fits one series with one analytic q, sampled mid-season.
- **Size-frequency data.** SS3's generalised size compositions for the survey ($\lambda = 0.51$) are not converted.
- **Plus-group equilibrium.** SS3 iterates it to 3 × Nages; opal uses the geometric tail. This makes a difference of about 5e-5 in SPR0.
- **Selectivity (fits C and C2).** opal's curves are length-based and map to age through the PLA; SS3's are age-based. Both use the same growth here, so the difference is in curve shape, not in the age-length key.

11 Credit

The baseline SS3 model (`model-files/ss3/xx-io-yft-sim-iattc-sample-tree-4`) was developed by the IATTC analyst team for the Spatial Stock Assessment Simulation Experiment: Mark N. Maunder, Haikun Xu and Carolina Minte-Vera (Inter-American Tropical Tuna Commission). The yellowfin assessments from the experiment, including theirs, are evaluated in Goethel, D.R., et al. (2024). 'Drivin' with your eyes closed': Results from an international, blinded simulation experiment to evaluate spatial stock assessments. *Fish and Fisheries* 25(3): 471–490. https://doi.org/10.1111/faf.12819

The age-specific K growth model is Stock Synthesis' `GrowthModel 3` (Methot, R.D. Jr., and Wetzel, C.R. 2013. Stock synthesis: A biological and statistical framework for fish stock assessment and fishery management. *Fisheries Research* 142: 86–99. https://doi.org/10.1016/j.fishres.2012.10.012; source: https://github.com/nmfs-ost/ss3-source-code). It is implemented in opal from version 0.1.1.

The data and the simulated truth come from replicate 4 of the Indian Ocean yellowfin operating model of the Spatial Stock Assessment Simulation Experiment (repository https://github.com/aaronmberger-nwfsc/Spatial-Assessment-Modeling-Workshop). That operating model was built in SPM (Dunn, Rasmussen and Mormede 2015), based on Dunn, Hoyle and Datta (2020), with growth from Fu et al. (2018) after Fonteneau and Gascuel (2008). See `model-files/sim-truth/io-yft-iattc-sample-tree-4/README.md` for file provenance.

12 Session

item	value
r	4.5.3
opal	0.1.1
r4ss	1.56.0
cpu	Intel(R) Xeon(R) Processor @ 2.80GHz
cores	4